

Improving Explainable Chart Understanding in Vision-Language Models

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I. MOTIVATION

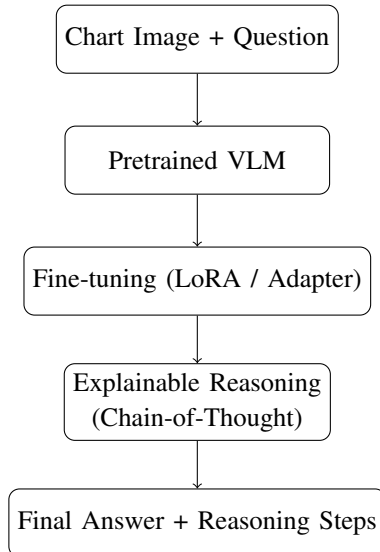
Visual-Language Models (VLMs) have demonstrated strong performance in general multimodal reasoning. However, they sometimes struggle with structured visual reasoning tasks, particularly understanding data visualizations such as bar charts, scatterplots, and heatmaps. These charts require precise visual perception, numeric extraction, and logical reasoning, which differ from natural image understanding.

Furthermore, VLM predictions are often opaque, making it unclear whether the model truly understands the visualization. This project investigates whether **Chain-of-Thought (CoT) reasoning** can improve both the accuracy and interpretability of VLMs when answering questions about charts using the ChartQA dataset.

II. PROJECT OVERVIEW

We propose to fine-tune a pretrained Vision-Language Model on the ChartQA dataset and introduce an explainable reasoning mechanism in which the model generates intermediate reasoning steps before producing a final answer. The goal is to improve:

- Numerical accuracy
- Logical reasoning over charts
- Interpretability of predictions



III. METHODOLOGY

We conduct an experiment to evaluate whether explainable reasoning improves multimodal chart understanding.

Task Setup. Given a chart image and a natural language question, the model predicts a short textual answer.

- Input: chart image, question
- Output: answer (numeric or short phrase), optionally with reasoning steps.

Baseline. We first evaluate a pretrained Vision-Language Model (VLM) using direct answering without reasoning on the ChartQA validation set.

Fine-tuning. We apply lightweight fine-tuning (LoRA or partial parameter tuning) on the ChartQA training set to adapt the VLM to chart-specific visual and numerical reasoning.

Explainable Reasoning (Chain-of-Thought). We introduce reasoning-enabled prompting where the model generates intermediate reasoning steps, such as identifying axes, extracting relevant values, and performing logical or numerical computation, before producing the final answer.

Comparison. We compare three configurations:

- Zero-shot baseline vs fine-tuned model
- Direct answering vs Chain-of-Thought reasoning
- Accuracy vs interpretability trade-off

IV. EVALUATION STRATEGY

We evaluate on the ChartQA using two metrics:

Quantitative Metrics

- Exact Match (EM) accuracy
- Numeric Accuracy (with tolerance for floating-point answers)

Qualitative Analysis

- Compare reasoning-enabled vs non-reasoning predictions
- Error analysis (numeric reasoning, axis misinterpretation, logical errors)
- Case studies demonstrating improved interpretability